Raw Data

Masked Language Models and BERT

Cornell CS 5740: Natural Language Processing
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Masked LMs

- LMs so far: predict the next token given the previous tokens
- This enables a self-supervised task
- That we can train on a lot of data
- And get really interesting and useful representations

Masked LMs

- LMs so far: predict the next token given the previous tokens
- What if we have a complete sentence?
  - Can do the same

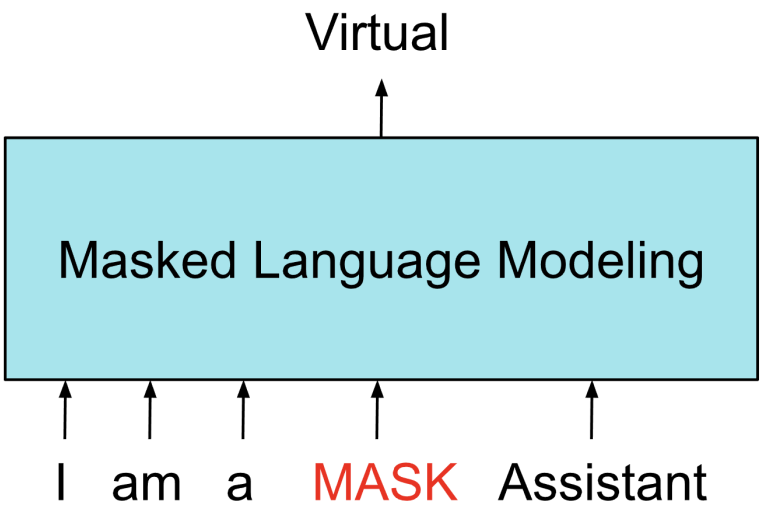
Masked LMs

- LMs so far: predict the next token given the previous tokens
- What if we have a complete sentence?
  - Can do the same
  - Decode through an LM to compute representations
  - But: representations conditioned on past context only
  - So: missing an opportunity to incorporate future context
- How can we formulate a self-supervised prediction task?

Masked LMs

- We have many sequences of tokens $\bar{x} = \langle x_1, \ldots, x_n \rangle$
  - Just raw data, like with regular LMs
- Let's create a prediction task by hiding part of the sequence, and then trying to predict them
  - Input: the sequence $\bar{x}^M$ where some tokens are replaced with the token $[\mathrm{MASK}]$, for example: $\bar{x}^M = \langle x_1, \ldots, x_4, [\mathrm{MASK}], x_6, \ldots, x_n \rangle$
  - Output: a probability distribution over tokens for each of the masked tokens
  - Goal: for each masked position, the correct token gets the highest probability, such that, for example, $\arg\max_{\mathscr{V}} p(x_7^M \mid \bar{x}^M) = x_7$
  - Training objective: negative log-likelihood for masked tokens

Masked LMs



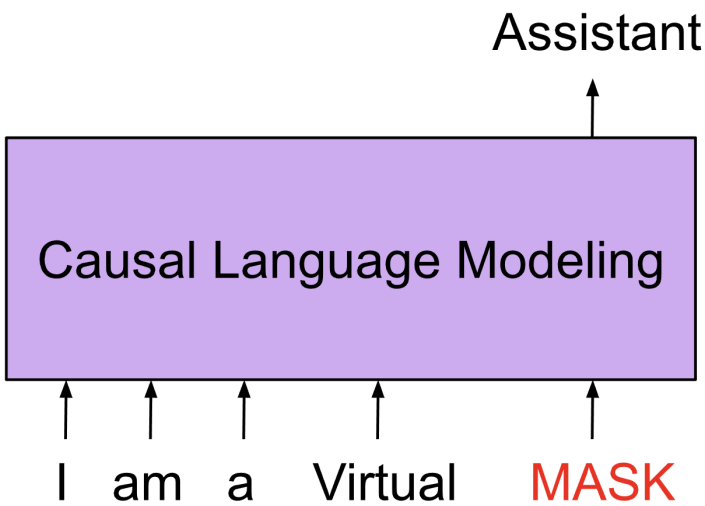


Image from https://www.holisticai.com/blog/from-transformer-architecture-to-prompt-engineering

Encoder Transformer

- So far: Transformers self-attend to past tokens to predict the next token
  - This is called a Decoder Transformer
- Encoders assume we have the complete sequence
- There is no generation problem, we just want representations
  - We will learn how to use them later on
- The big difference: self-attention is not masked, so computes weighted sum over entire context (i.e., entire sequence)

The Transformer

Decoder-only Variant (revisit)

$\text{TransformerBlock}^{\text{k}}(\mathbf{u}_1, \ldots, \mathbf{u}_{\text{i}})$

$$\mathbf{q}^{(l)} = \text{Linear}_{\text{q}}^{(\text{l})}(\mathbf{u}_{\text{i}})$$

$$\mathbf{K}^{(l)} = \text{Linear}_{\text{k}}^{(\text{l})}([\mathbf{u}_1 \cdots \mathbf{u}_{\text{i}}])$$

$$\mathbf{V}^{(l)} = \text{Linear}_{\text{v}}^{(\text{l})}([\mathbf{u}_1 \cdots \mathbf{u}_{\text{i}}])$$

$$\mathbf{z} = \text{LN}(\text{Linear}_{\text{z}}([\text{SelfAttn}(\mathbf{q}^{(1)}, \mathbf{K}^{(1)}, \mathbf{V}^{(1)}); \cdots; \text{SelfAttn}(\mathbf{q}^{(L)}, \mathbf{K}^{(L)}, \mathbf{V}^{(L)})]) + \mathbf{u}_i)$$

$$\mathbf{h}_i^k = \text{LN}(\text{Linear}_2(\text{GELU}(\text{Linear}_1(\mathbf{z}))) + \mathbf{z})$$

$$\mathbf{x}_i = \phi(x_i) + \phi_p(i)$$

$$\mathbf{h}_i^1 = \text{TransformerBlock}^1(\mathbf{x}_1, \ldots, \mathbf{x}_{\text{i}})$$

$$\mathbf{h}_i^2 = \text{TransformerBlock}^2(\mathbf{h}_1^1, \ldots, \mathbf{h}_{\text{i}}^1)$$

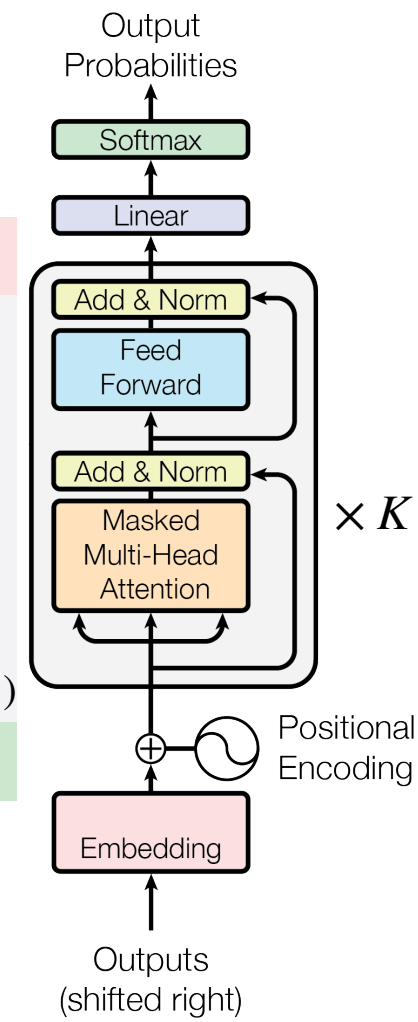
$$\ldots$$

$$\mathbf{h}_i^k = \text{TransformerBlock}^{\text{k}}(\mathbf{h}_1^{\text{k}-1}, \ldots, \mathbf{h}_{\text{i}}^{\text{k}-1})$$

$$\ldots$$

$$\mathbf{h}_i^K = \text{TransformerBlock}^{\text{K}}(\mathbf{h}_1^{\text{K}-1}, \ldots, \mathbf{h}_{\text{i}}^{\text{K}-1})$$

$$p(x_{i+1} \mid x_1, \ldots, x_i) = \text{softmax}(\text{Linear}_{\mathscr{V}}(\mathbf{h}_{\text{i}}^{\text{K}}))$$



Self-attention reminder

$$\text{SelfAttn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}(\mathbf{QK}/\sqrt{d_k})\mathbf{V}$$

During learning, compute the whole sequence at ones by **masking** items you shouldn't attend to in softmax — easy by setting softmax to $-\infty$

[Vaswani et al. 2017]

Encoder Transformer*

$\text{TransformerBlock}^{\text{k}}(\mathbf{u}_1, \ldots, \mathbf{u}_\text{n})$

$$\mathbf{Q}^{(l)} = \text{Linear}_\text{q}^{(\text{l})}([\mathbf{u}_1 \cdots \mathbf{u}_\text{n}])$$

$$\mathbf{K}^{(l)} = \text{Linear}_\text{k}^{(\text{l})}([\mathbf{u}_1 \cdots \mathbf{u}_\text{n}])$$

$$\mathbf{V}^{(l)} = \text{Linear}_\text{v}^{(\text{l})}([\mathbf{u}_1 \cdots \mathbf{u}_\text{n}])$$

$$\mathbf{Z} = \text{LN}(\text{Linear}_\text{z}([\text{SelfAttn}(\mathbf{Q}^{(1)}, \mathbf{K}^{(1)}, \mathbf{V}^{(1)}); \cdots; \text{SelfAttn}(\mathbf{Q}^{(L)}, \mathbf{K}^{(L)}, \mathbf{V}^{(L)})]) + [\mathbf{u}_1 \cdots \mathbf{u}_n])$$

$$[\mathbf{h}_1^k \cdots \mathbf{h}_n^k] = \text{LN}(\text{Linear}_2(\text{GELU}(\text{Linear}_1(\mathbf{Z}))) + \mathbf{Z})$$

$$\mathbf{x}_i = \phi(x_i^M) + \phi_p(i), i = 1, \ldots, n$$

$$[\mathbf{h}_1^1 \cdots \mathbf{h}_n^1] = \text{TransformerBlock}^1(\mathbf{x}_1, \ldots, \mathbf{x}_\text{n})$$

$$[\mathbf{h}_1^2 \cdots \mathbf{h}_n^2] = \text{TransformerBlock}^2(\mathbf{h}_1^1, \ldots, \mathbf{h}_\text{n}^1)$$

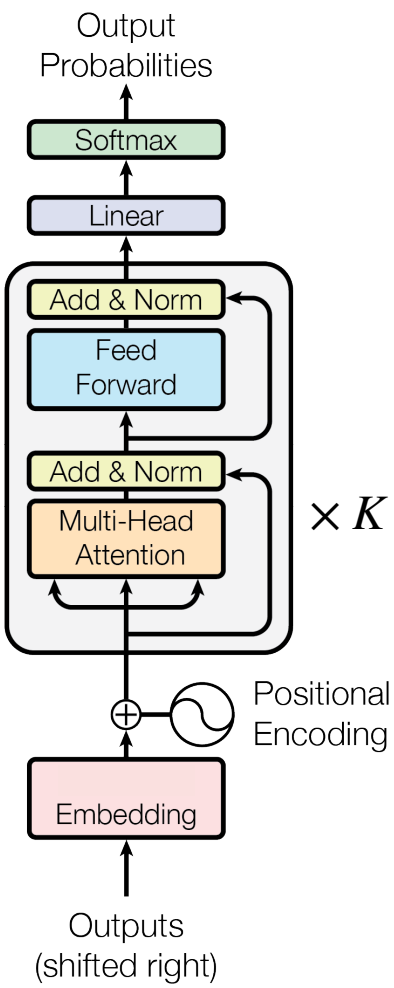
$$\ldots$$

$$[\mathbf{h}_1^k \cdots \mathbf{h}_n^k] = \text{TransformerBlock}^\text{k}(\mathbf{h}_1^{\text{k}-1}, \ldots, \mathbf{h}_\text{n}^{\text{k}-1})$$

$$\ldots$$

$$[\mathbf{h}_1^K \cdots \mathbf{h}_n^K] = \text{TransformerBlock}^\text{K}(\mathbf{h}_1^{\text{K}-1}, \ldots, \mathbf{h}_\text{n}^{\text{K}-1})$$

$$p(x_i \mid x_1^M, \ldots, x_n^M) = \text{softmax}(\text{Linear}_{\mathcal{V}}(\mathbf{h}_\text{i}^\text{K}))$$



Self-attention reminder

$$\text{SelfAttn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}(\mathbf{QK}/\sqrt{d_k})\mathbf{V}$$

* for Masked LM

[Vaswani et al. 2017]

BERT

Bidirectional Encoder Representations from Transformers

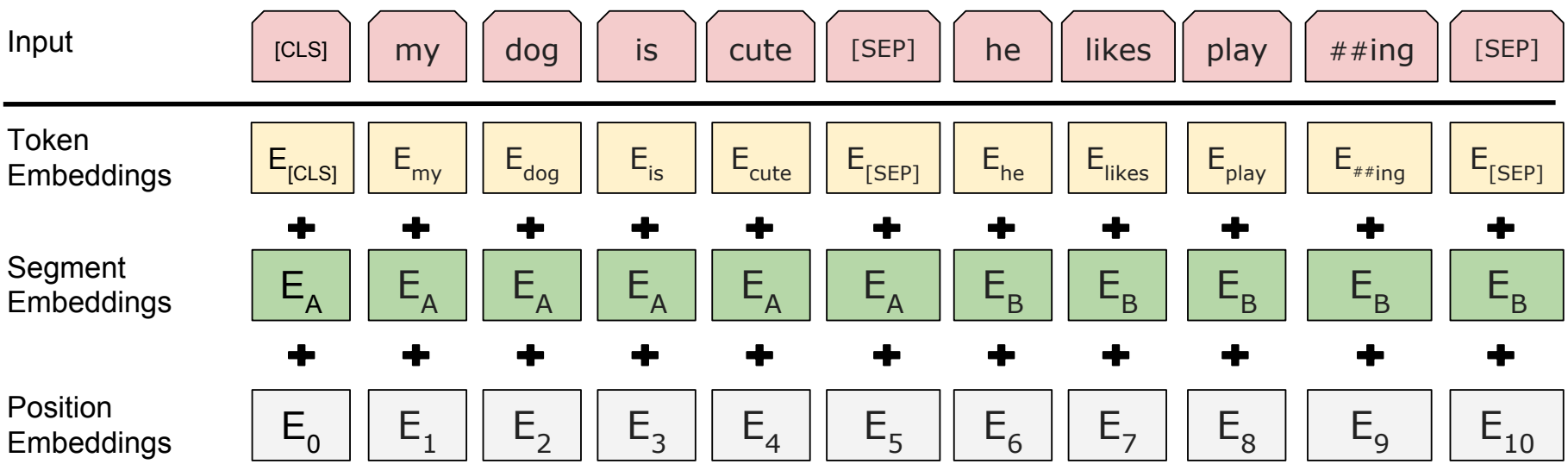
- Encoder transformer
- BERT Base: 12 transformer blocks, 768-dim word-piece tokens, 12 self-attention heads → 110M parameters
- BERT Large: 24 transformer blocks, 1024-dim word-piece tokens, 16 self-attention heads → 340M parameters
- RoBERTa: same model, much more data (160GB of data instead of 16GB)

[Devlin et al. 2018]

BERT

Inputs

- One or two sentences
  - Word-piece token embeddings
  - Position and segment embeddings



[figure from Devlin et al. 2018]

BERT

Word-piece Tokenization (in a nutshell)

1. Initialize with tokens for all characters
2. While vocabulary size is below the target size:
    1. Build a language model over the corpus (e.g., unigram language model)
    2. Merge pieces that lead to highest improvement in language model perplexity

- Need to choose a language model that will make the process tractable
- Often a unigram language model (e.g., SentencePiece library)
- Particularly suitable for machine translation

[Schuster and Nakajima (2012), Wu et al. (2016), Kudo and Richardson (2018)]

BERT

Training

- Data: raw text
- Two objectives:
  - Masked LM
  - Next-sentence prediction
- No attention mask — why?
- Later development in RoBERTa:
  - More data, no next-sentence prediction, dynamic masking

BERT

Masking Recipe for Training

- Mask and predict 15% of the tokens
  - For 80% (of 15%) replace with the input token with $[\mathrm{MASK}]$
  - For 10%, replace with a random token
  - For 10%, keep the same

BERT

Next-sentence Prediction

- Input: [CLS] Text chunk 1 [SEP] Text chunk 2
- Training data: 50% of the time, take the true next chunk of text, 50% of the time take a random other chunk
- Predict whether the next chunk is the true next chunk
- Prediction is done on the [CLS] output representation

BERT

Related Techniques

- Central Word Prediction Objective (context2vec) [Melamud et al. 2016]
- Machine Translation Objective (CoVe) [McMann et al. 2017]
- Bi-directional Language Modeling Objective (ELMo) [Peters et al. 2018]
- Then BERT came …
- … and many more followed

BERT

What Do We Get?

- We can feed complete sentences to BERT
- For each token, we get a contextualized representation
  - Meaning: computed taking the other tokens in the sentence into account
- In contrast to word2vec representations that are fixed and do not depend on context
- While word2vec vectors are forced to mix multiple senses, BERT computes instance-specific vectors

BERT

Learned Representations

- Given a sequence of tokens $\bar{x} = \langle x_1, \ldots, x_n \rangle$, BERT computes
  - Hidden representations for each layer $\mathbf{h}_i^k, i = 1,\ldots,n \quad k = 1,\ldots,K$
- Each hidden representation is computed from all tokens in the input
- Traditionally, the CLS token representation represents the entire input
- Each token has hidden representations $\mathbf{h}_i^1, \ldots, \mathbf{h}_i^K$
- Different layers $k = 1,\ldots,K$ may represent different aspects of meaning
- These representations (or combination of them) can be used similar to word/ sentence embeddings
  - Note: sub-word tokenization vs. words — why does this matter?

BERT

How Do We Use It?

- Widely supported by existing frameworks
  - E.g., Transformers library by Hugging Face
- There are specific recipes on how to use — soon, when we discuss working with annotated data
- Large BERT models quickly outperformed human performance on several NLP tasks
  - But what it meant beyond benchmarking was less clear
- Started an arms race towards bigger and bigger models, which quickly led to the LLMs of today

BERT

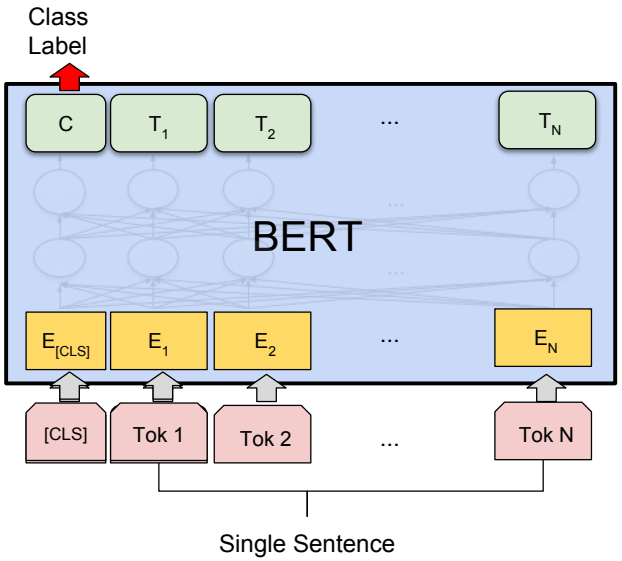
For Classification

- Given a sentence $\bar{x} = \langle x_1, \ldots, x_n \rangle$, add CLS token
- Compute representations using BERT, and classify using the CLS representation

$$\mathbf{h}_{\text{CLS}}, \mathbf{h}_1, \ldots, \mathbf{h}_n = \text{BERT}(\text{CLS}, x_1, \ldots, x_n)$$

$$p(\hat{y} \mid \bar{x}) = \text{softmax}(\mathbf{h}_{\text{CLS}}\mathbf{W})$$

- Training:
  - Fine-tuning all BERT weights
  - Following BERT best practices



BERT

Usage Best Practices

- Use the pre-trained model as the first “layer” of your final model
- Train with fine-tuning using your supervised data
- Fine-tuning recipe: 1-3 epochs, batch size 2-32, learning rate 2e-5 - 5e-5
  - Large changes to weights in top layers (particularly in last layer to route the right information to [CLS])
  - Smaller changes to weights lower down in the transformer
  - Small learning rate and short fine-tuning schedule mean weights don’t change much
  - More complex recipes exist, but often not necessary (see Zhang et al. 2021 for study of stability and good practices)

BERT

What It Is Not Great For?

- BERT cannot generate text (at least not in an obvious way)
  - Not an autoregressive model, can do weird things like stick a [MASK]at the end of a string, fill in the mask, and repeat
- Masked language models are intended to be used primarily for “analysis” tasks

BERT

What does BERT Learn?

- There is a lot of work trying to decipher what BERT learns in its representations
  - Much harder with recent LLMs because they are not as open
- Some very interesting results, but not completely clear how to interpret them

What Does BERT Learn?

- Try to solve different linguistic tasks given each block level, without fine-tuning
  - Specifically: solve tasks using mixing weights on levels
- Goal: see what information each new level adds
- Each task classifier takes a single mixed hidden representation $\mathbf{h}_{i,\tau}$ or a pair of representations for two tokens

i : token index

K : number of block levels

τ : task

γ_τ : task parameter

$\mathbf{a}_\tau$: mixing parameters

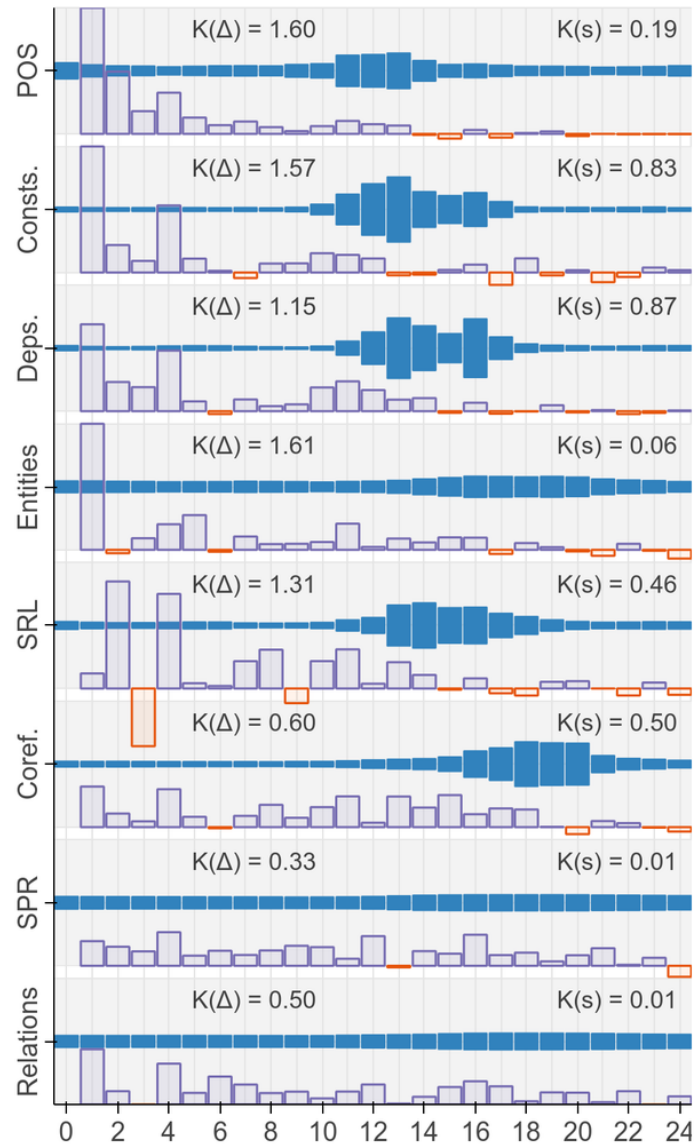
$$\mathbf{s}_\tau = \mathrm{softmax}(\mathbf{a}_\tau)$$

$$\mathbf{h}_{i,\tau} = \gamma_\tau \sum_{k=0}^{K} s_\tau^k \mathbf{h}_i^k$$

[Tenney et al. 2019]

What Does BERT Learn?

- Each plot shows a task
- Plots show $\mathbf{s}_\tau^k$ weights magnitude in blue, and the number of self-attention levels
- The performance delta when adding this layer is in purple
- Largely: higher level semantic tasks happen in later levels



[figure from Tenney et al. (2019)]

Extra Readings

- BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding
- Deep contextualized word representations
- ViLT: Vision-and-Language Transformer Without Convolution or Region Supervision

Acknowledgements

We thank the following sources for the materials on which slides on this deck are based:

- UT Austin CS 388 by Greg Durrett